

Effective Apodization in Synthetic Aperture Ultrasound Computed Tomography

Soheil Hakakzadeh

Research and Development
Med Farasout Tajhiz Co.,
Sharif University of Technology
Tehran, Iran
Soheilh@ieee.org

Zahra Kavehvash

Research and Development
Med Farasout Tajhiz Co.,
Sharif University of Technology
Tehran, Iran
Kavehvash@sharif.edu

Mohammad Mehrmohammadi

Department of Imaging Science
University of Rochester Medical Center
Rochester, NY, USA
mohammad_mehr@urmc.rochester.edu

Abstract— This paper presents a novel apodization technique designed to enhance image quality in synthetic aperture reflection ultrasound computed tomography (SA-RUCT) systems. Conventional SA-RUCT methods often suffer from artifacts and reduced image quality, particularly when using a large number of receiving elements (Rxs). These issues arise due to diffraction and distortion effects, which degrade the signal-to-noise ratio (SNR) and overall image clarity. The proposed method addresses these challenges by optimizing the apodization process, leading to significant reductions in artifacts and improvements in both resolution and contrast. Numerical simulations indicated that the proposed method makes a significant improvement in peak signal-to-noise ratio (PSNR) compared to conventional methods. Experimental and *in-vivo* studies further validated these findings, showing enhanced visualization of fine structures and effective suppression of unwanted artifacts. The proposed method achieved spatial resolution improvements of up to nearly 60%, in the reconstructed images. The results indicate that the proposed apodization technique is a robust and practical solution for improving image quality in SA-RUCT systems, offering significant advantages in clinical and research applications where high-resolution and artifact-free imaging is critical.

Keywords— Reflection Ultrasound Computed Tomography, Apodization, Synthetic Aperture, Image reconstruction.

I. INTRODUCTION

Ultrasound Computed Tomography (USCT) is an advanced imaging technique that has gained significant attention, particularly for breast cancer diagnosis [1]. Unlike conventional ultrasound (i.e. planar), which relies on reflected sound waves, USCT captures the complete ultrasonic wavefield transmitted through the object [2]. This method uses multiple transducers arranged around the imaging target to reconstruct cross-sectional images, offering detailed insights into tissue structure [3, 4].

USCT's primary advantage is its ability to produce images with higher spatial resolution and contrast, crucial for detecting small lesions that may go unnoticed with other modalities [5]. In a USCT system, synthetic aperture reflection ultrasound computed tomography (SA-UCT) enhances resolution and contrast of the reconstructed image by employing a synthetic aperture approach [6]. In SA-UCT, each transducer emits an ultrasonic pulse sequentially, with others acting as receivers. This process creates a comprehensive dataset of reflected echoes, allowing high-resolution image reconstruction by combining multiple perspectives, similar to synthetic aperture radar (SAR).

The quality of reconstructed images in SA-UCT depends on several factors, including the number of transmitting (Tx) and receiving (Rx) elements [7]. More Tx elements provide comprehensive coverage, while increasing Rxs enhances SNR by collecting more data from reflected echoes. However, using large number of Rxs can lead to diffraction and distortion, appearing artifacts that obscure critical details. Therefore, optimizing number of Tx and Rx elements is crucial for high-quality imaging [6].

Research studies have been focused on improving the quality of reconstructed images. Advanced beamforming techniques, such as delay-and-sum (DAS) and synthetic aperture focusing, have been employed to enhance spatial resolution [8, 9]. Adaptive beamforming methods such as minimum variance [10] and weighting factors [11, 12] have also been proposed to reduce artifacts and improve image clarity, particularly in complex tissues. Novel apodization and wavelet processing methods have been proposed to mitigate diffraction and side-lobe artifacts in USCT systems [13-15]. Advanced transmission techniques, such as diverging waves [16] and phased-array compounding [17], are also introduced to enhance the signal-to-noise ratio (SNR). Additionally, hybrid imaging approaches, combining RUCT with other modalities like photoacoustic imaging, have been explored to improve diagnostic accuracy [18-21]. These multimodal techniques leverage the strengths of each modality for better tissue characterization and abnormality detection.

The present research aims to introduce an innovative, low-cost apodization technique to improve image quality in SA-UCT systems. By applying a weighting map based on the angle between Tx and Rx rays for each Tx-Rx pair, the proposed method aims to reduce artifacts and enhance the resolution and contrast of pulse-echo images, optimizing overall system performance.

II. MATERIALS AND METHODS

In SA-UCT, the quality of the reconstructed image heavily depends on the number of Rx elements in the array. As the number of Rx elements increases, the SNR value of the reconstructed image generally improves, leading to clearer and more detailed imaging. This is because more Rx elements capture additional reflected signals from different angles, providing a richer dataset that enhances the accuracy of the image reconstruction process. However, this improvement comes with trade-offs. When too many receivers are involved, diffraction and distortion effects become more pronounced, particularly at off-center distances. These effects can

introduce artifacts and cause image blurring, reducing its overall quality. Conversely, if the number of Rx elements is too low, the SNR drops significantly, resulting in images that may be too noisy to be diagnostically useful. Thus, there is a critical balance to be struck in determining the optimal number of receiving elements to maximize image quality while minimizing artifacts.

A. Artifact Analysis

The performance of the SA-UCT system was evaluated through a numerical study that focused on the SNR and the presence of artifacts in the reconstructed images. The study was conducted using the k-Wave MATLAB toolbox [22], a numerical tool designed for modeling acoustic wave fields. The experimental setup consisted of a circular array with a 40 mm radius and 128 Tx elements. The number of Rx elements varied between 1 and 128 to study their impact on image quality.

The images in Fig. 1(a) illustrate the imaging object—a phantom consisting of three closely spaced circular inclusions. Figs. 1(b1-b3) show the reconstructed images using different numbers of Rx elements: 1, 21, and 61, respectively. As expected, increasing the number of Rx elements improved the SNR due to the higher volume of reflected signals captured. This enhancement is evident when comparing the images: the image reconstructed with a single Rx element [Fig. 1(b1)] exhibits significant noise, obscuring the phantom's structure. As the number of Rx elements increased to 61 [Fig. 1(b3)], the image quality improved, but some artifacts persisted, particularly in regions farther from the transducers. Finally, using 21 Rx elements [Fig. 1(b2)] resulted in a much clearer image with reduced noise and artifacts, allowing for a more accurate representation of the phantom.

Despite the overall improvement in SNR, the images shown in Figure 1 highlight the challenge of diffraction and distortion artifacts that become more pronounced as the number of Rx elements increases. These artifacts can obscure critical details, particularly in complex tissue environments, underscoring the need for optimized apodization techniques to balance SNR enhancement and artifact minimization.

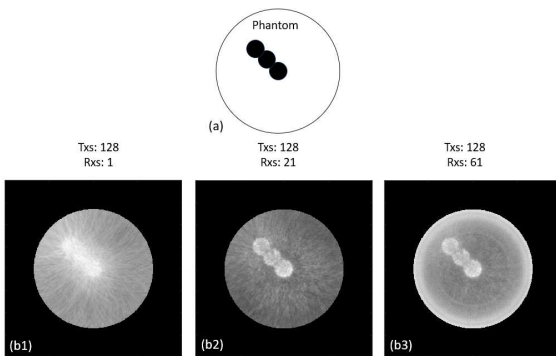


Fig. 1: Numerical study illustrating the impact of the number of Rx elements on the reconstructed image quality in a SA-UCT system. (a) Diagram of the imaging phantom, consisting of three closely spaced circular inclusions. (b1) Reconstructed image using 1 Rx element, showing significant noise and poor image quality. (b2) Reconstructed image using 21 Rx elements, with improved SNR but noticeable artifacts. (b3) Reconstructed image using 61 Rx elements, demonstrating further improvement in image clarity and reduced artifacts, providing a clearer representation of the phantom structure.

B. Apodization Technique

Based on the previously presented data, as the number of Rx elements in SA-UCT increases, the SNR improves, but this also leads to more pronounced diffraction and distortion artifacts. These artifacts degrade the quality of the reconstructed images, particularly in regions where the acoustic waves experience significant scattering and interference.

The nature of these artifacts can be explained by the Rayleigh scattering model [23], which describes how the scattering of acoustic waves is influenced by the angle between the Tx and Rx elements. According to this model, the intensity of scattered waves varies with the angle, causing different levels of distortion depending on the relative positions of the Tx and Rx pairs. Specifically, at certain angles, the scattering is more intense, leading to greater artifacts in the reconstructed images.

To address this issue, we developed a novel apodization technique based on the Rayleigh scattering model. The proposed method involves applying an angular weighting function that accounts for the varying intensity of scattered waves at different angles. This apodization function is designed to reduce the contributions of signals received at angles where scattering is most pronounced, thereby minimizing the associated artifacts.

The weighting function used in this study is derived from the Rayleigh scattering model and is defined as [23]:

$$R(\theta) = \frac{1}{r^2} \left(1 - \frac{3 \cos(\theta)}{2} \right)^2 \quad (1)$$

In this equation, θ represents the angle between the Tx-Rx pair, and r is the distance between the transmitting and the reflecting particle. The function $R(\theta)$ effectively reduces the amplitude of signals received at angles where the scattering leads to significant artifacts.

This apodization map was applied to the pulse-echo data from each Tx-Rx pair before reconstruction. The final image was obtained by multiplying the pulse-echo map with the apodization map, which led to a significant reduction in artifacts and an improvement in image clarity, particularly in the areas where traditional reconstructions exhibited pronounced distortions.

The effectiveness of the proposed apodization technique is evaluated through a series of numerical simulations, experimental setups with tissue-mimicking phantoms, and *in-vivo* studies. The results of these evaluations are presented in the following sections, where the performance of the method in reducing artifacts and enhancing image quality is thoroughly analyzed.

III. RESULTS

A. Numerical Simulation Results

Numerical simulations were performed using the k-Wave MATLAB toolbox to assess the effectiveness of the proposed apodization technique in SA-UCT. The setup involved 512 transducers (TRs) arranged around a 40 mm radius region, with images reconstructed using 256 Rx elements.

As shown in Fig. 2, the results with only one receiving element ($N_{Rx} = 1$) using the conventional method [Fig. 2(a1)] displayed significant artifacts and distortions, leading to a PSNR of 25 dB. Increasing the number of Rxs to 256 ($N_{Rx} = 256$) improved the PSNR to 29 dB [Fig. 2(a2)], but also introduced additional artifacts, particularly near the transducers.

The proposed apodization method [Fig. 2(a3)] significantly reduced these artifacts and improved overall image quality, achieving a PSNR of 45 dB. This method demonstrated superior performance, particularly in suppressing bright ring artifacts and minimizing diffraction distortions, which are common in conventional methods.

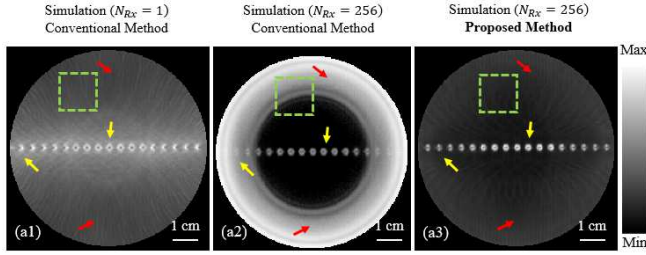


Fig. 2: Simulation results for different methods and receiving elements (N_{Rx}): (a1) Conventional method with $N_{Rx} = 1$ (PSNR = 25 dB), (a2) Conventional method with $N_{Rx} = 256$ (PSNR = 29 dB), and (a3) Proposed method with $N_{Rx} = 256$ (PSNR = 45 dB). The proposed method significantly reduces artifacts and improves image quality.

B. Experimental and In-Vivo Results

To further validate the proposed method, the proposed method was tested on both experimental phantom and *in-vivo* public dataset. The experimental setup consisted of a ring array with a radius of 40 mm. A total of 512 TRs and 256 Rxs, each with a central frequency of 5 MHz and a 60% bandwidth, were utilized in the image reconstruction process. The setup included a phantom composed of eight copper wires, each with a diameter of 0.1 mm, positioned perpendicular to the imaging plane, as well as an eight-week-old female athymic nude mouse for *in-vivo* studies [6].

The experimental results [Figs. 3(a1) and 3(a2)] revealed that the proposed apodization method effectively suppressed unwanted artifacts (highlighted by red and yellow arrows), which were prevalent in the conventional method. This improvement is evident in the sharper and more distinct features in the reconstructed images. Similarly, the *in-vivo* results [Figs. 3(b1) and 3(b2)] demonstrated enhanced image clarity and contrast, particularly in tissue boundaries and internal structures, when using the proposed method. The resolution of fine structures, such as the tissue boundaries and internal details, was significantly improved, with the proposed method outperforming the conventional approach in both resolution and contrast metrics.

Overall, the proposed apodization technique consistently outperformed the conventional method in both numerical simulations and real-world studies, proving its effectiveness in improving image quality and reducing artifacts in SA-UCT systems.

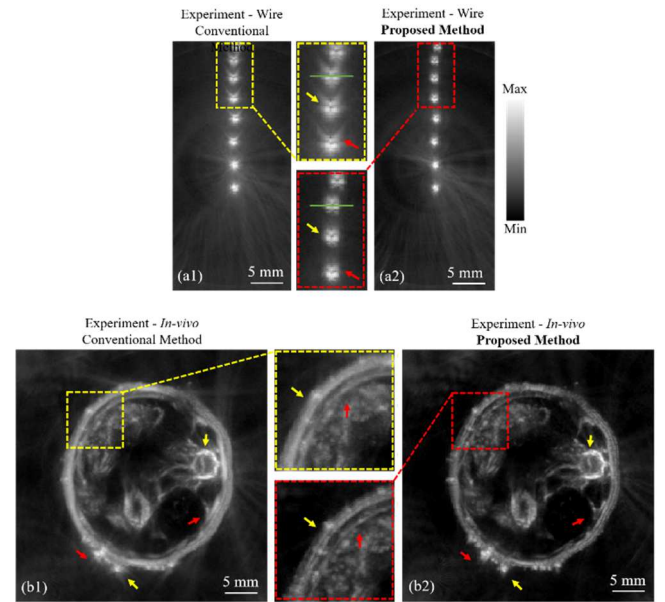


Fig. 3: Comparison of experimental and *in-vivo* imaging results using the conventional method and the proposed apodization technique. (a1, a2) Experimental images show artifact suppression and improved feature clarity with the proposed method. (b1, b2) *In-vivo* images demonstrate enhanced resolution and contrast, particularly in tissue boundaries and internal structures, when using the proposed technique. Yellow and red arrows highlight areas of significant improvement.

IV. DISCUSSION AND CONCLUSION

The results of this study demonstrate that the proposed apodization technique significantly enhances the quality of reconstructed images in SA-RUCT. Through both numerical simulations and experimental validations, it was evident that the conventional method, suffers from artifacts and reduced image quality as the number of Rx elements increases. These issues were especially pronounced near the transducers due to diffraction and distortion effects, which degrade the SNR and overall image clarity.

The proposed method effectively addresses these challenges by optimizing the apodization process, thereby reducing artifacts and enhancing the resolution and contrast of the images. In numerical simulations, the proposed method achieved a lateral resolution improvement of up to 60% dB compared to the conventional approach (see dashed-lines in Figs. 3(a1) and 3(a2)). Experimental and *in-vivo* studies further validated these findings, showing clear improvements in the visualization of fine structures and suppression of unwanted artifacts.

The implications of these findings are significant for the field of medical imaging, particularly in applications where high-resolution and artifact-free images are critical. The ability to achieve such improvements without increasing the complexity of the imaging system makes this technique highly practical and scalable.

In conclusion, the proposed apodization technique offers a robust solution for improving image quality in SA-RUCT systems. By addressing the inherent limitations of conventional methods, this approach opens new possibilities for more accurate and reliable imaging in clinical settings. Future work will focus on refining the technique and exploring its application in a broader range

of imaging scenarios, including more complex *in-vivo* studies and real-time imaging environments.

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